Evaluating Reciprocity Norms and Emotional Comprehension in LLM Decision-Making

Emotional Framing and Reciprocity in Economic Games: A Computational Neuroeconomic Study

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1. Problem Statement

In humans, fairness and reciprocity are not purely calculative. Unfair Ultimatum offers elicit greater anterior-insula activity, with stronger activity associated with rejection (Sanfey et al., 2003). Disrupting the right DLPFC (Knoch et al., 2006), serotonin depletion (Crockett et al., 2008), oxytocin (Kosfeld et al., 2005) and incidental sadness (Harlé & Sanfey, 2007) can alter social economic decisions. Communicated anger can also alter bargaining concessions, with effects depending on the strategic structure of the interaction (van Dijk et al., 2008). LLMs are increasingly studied as economic agents and simulated participants (Horton, 2023; Aher et al., 2023; Mei et al., 2024), and recent work shows that emotional states induced *in the model itself* alter its play in strategic games (Mozikov et al., 2024). What remains less explored is the complementary, interpersonal case: systematic manipulation of a counterpart's *expressed* emotion as a moderator of reciprocity, with economic payoffs held constant.

Research question: Does emotional information about a counterpart systematically change an LLM's economic decisions when the payoff structure is held constant, and does the direction of that change resemble patterns reported in human neuroeconomic research?

Positioning. We do not claim that LLMs "feel" emotions or that we measure neural activity. In Marr's terms, neuroeconomics provides evidence about the implementation of affective systems in human social decision-making. We instead test for an analogous behavioural pattern: emotion-conditioned shifts in fairness thresholds and reciprocity in an artificial decision-maker without human neural substrates. A positive result means only that decisions are sensitive to emotionally framed social information; candidate explanations (learned social norms, instruction tuning/RLHF, contextual inference) are discussed but not inferred.

2. Hypotheses

- **H1 – Negative reciprocity:** In the Ultimatum Game (UG), acceptance increases with offer size, with a positive minimum acceptable offer (MAO), implying costly rejection of sufficiently low offers.
- **H2 – Positive reciprocity:** In the Trust Game (TG), the share returned increases with the amount sent by the investor.
- **H3 – Emotion beyond incentives (core):** At a fixed economic offer, counterpart emotion shifts the decision. We expect an angry or entitled proposer to raise the MAO and a sad or apologetic proposer to lower it, although the human evidence is mixed (expressed anger can elicit either concession or reactance). Because anger and sadness share negative valence, opposite effects would suggest emotion-specific rather than purely valence-based effects. Happy is exploratory.
- **H4 – Strategic use of emotion:** Recipient anger should increase offers in UG, where the recipient can veto, but not in the Dictator Game (DG), where the recipient cannot. Sadness is expected to increase giving in both games. This is tested through an Emotion × Game interaction.
- **H5 – Cross-model robustness (exploratory):** effect direction and magnitude vary across model families.

3. Experimental Design

The three games separate different social motives. The DG provides a baseline for unilateral giving; the UG captures fairness, costly punishment and strategic giving, and comparing UG with DG giving helps examine the contribution of strategic incentives (Forsythe et al., 1994); the TG measures trust and positive reciprocity.

Emotion manipulation: Before the decision, the counterpart sends a short message conveying Neutral, Angry, Sad, or Happy affect through message content rather than an explicit label, so an emotion-recognition check is performed. Messages are length-matched, with five variants per emotion (stimulus sampling), and are screened so that none states an intention to accept, reject or retaliate. Payoffs are identical across emotion conditions.

Sub-experiment	Manipulated factors	Cells
S1 UG–responder	Proposer emotion (4) × Offer (10/20/30/40/50% of pie) × Stakes (INR 100 / INR 10,000)	40
S2 UG vs DG–proposer	Recipient emotion (4) × Game (UG/DG) × Stakes (2); % of pie given	16
S3a TG–trustee	Investor emotion (4) × Amount sent (20/50/100% of endowment, tripled) × Stakes (2)	24
S3b TG–investor	Trustee emotion (4) × Stakes (2); % of endowment sent	8
Factorial cells per model		**88**
No-message baselines (identical payoff grids, counterpart message removed)	S1 +10, S2 +4, S3a +6, S3b +2	+22
Total cells per model		**110**

Controls: length-matched neutral messages; fresh context for every trial (single-shot, no memory); neutral role labels ("Player A/B", with no names or demographic cues); randomized response-option order; payoffs framed in INR; refusals and off-format outputs logged and reported, never silently dropped.

4. Methodology for Analysis

- **Models:** The core experiment will use two instruction-tuned LLMs from distinct model families. One will preferably be an open-weight 7–9B model run locally, with token-level log-probabilities where supported by the selected model/runtime, plus one closed API model. Exact model/version strings are pinned and logged. Temperature = 1.0, with a robustness check at 0.7 where feasible.
- **Sampling:** 4 instruction paraphrases × 4 message variants sampled from the five available variants = 16 prompt variants per cell, with 3 stochastic samples each. Because generalization is over prompts rather than samples, variants are prioritized over repeated draws. This yields 48 decisions/cell and 110 × 16 × 3 = 5,280 decisions/model, or 5,280 × 2 = 10,560 core decision calls across the two models. Output is forced to JSON; a one-line justification is requested on a random 20% subset.

- **Manipulation and belief checks:** Separate calls assess whether the intended emotion is recognized, its perceived intensity (1–7), perceived fairness where an allocation is present, and the perceived probability that the counterpart would reject or retaliate. The last item allows investigation of whether behavioural changes are accompanied by changes in beliefs about the counterpart's likely action. Checks are conducted on a sampled subset of prompt variants and cells rather than being repeated across the entire experiment. Recognition is summarized using confusion matrices.
- **Power and generalization:** Conventional human-subject power calculations do not apply directly, because repeated generations from one model are not independent participants. We therefore treat prompt variants and model identity as the units of generalization, and report effect sizes with CIs from the mixed models, which account for clustering within variants.
- **S1 analysis:** Mixed-effects logistic regression, reject ~ emotion × offer × stakes + (1|prompt variant). For each emotion, estimate the MAO (offer at which predicted P(accept) = 0.5) with bootstrap 95% CIs; the MAO shift is reported as percentage points of the pie associated with emotional framing. **Pre-registered fallback:** instruction-tuned models may accept (or reject) almost every offer, which would place the MAO outside the sampled range. If the pilot shows ceiling or floor effects we (i) extend the offer grid to 5% and 45%, and (ii) analyse the model's log-probability of accept as a graded dependent variable using the same regression, where supported, so that emotional modulation remains measurable even when the discrete choice does not move.
- **Computational model:** Fit Fehr–Schmidt (1999) disadvantageous inequity aversion with a softmax choice rule, P(accept) = sigma(lambda · [x - alpha · max(0, 1 - 2x)]), where x is the offered share. Estimating alpha by emotion tests whether emotional framing changes the fitted inequity-aversion parameter. Two caveats are pre-specified: alpha is only identified when the MAO falls inside the sampled offer range (it diverges as the MAO approaches 50%), and because x is a share, stake size cannot enter through alpha and is estimated in the regression instead.
- **S2/S3 analysis:** Linear mixed models on shares given, returned or sent (fractional/beta models if the distribution makes linear assumptions inappropriate). Planned emotion-vs-neutral contrasts use Holm correction; Emotion × Game tests H4 and Emotion × Amount-sent the relevant TG effects; model enters as a fixed factor for H5. Report odds ratios, percentage-point differences and Cohen's d with 95% CIs.
- **Qualitative analysis:** Both authors independently code a 10% subset of justifications for mentions of emotion, fairness and strategy, reporting Cohen's kappa.
- **Human comparison:** Baseline behaviour is compared with reported human meta-analytic benchmarks, not treated as a universal human standard: UG offers approx. 40% with approx. 16% rejection (Oosterbeek et al., 2004), DG giving approx. 28% (Engel, 2011), TG approx. 50% sent and approx. 37% returned (Johnson & Mislin, 2011). Comparisons concern broad patterns, not equality of effect sizes.

5. Plan, Scope, and Feasibility

The full factorial design (110 cells/model) is preserved so every hypothesis test keeps its intended resolution — in particular the 5-point offer grid for the MAO curve (H1/H3) and the Emotion × Game / Emotion × Amount-sent interactions (H4, H2). Feasibility comes instead from trimming sampling depth: 16 prompt variants (down from 20) × 3 stochastic samples (down from 5) yields **10,560 core decision calls** across the two models — roughly a 3× reduction from an unconstrained full design (~29–44k) — plus a sampled-subset of manipulation/belief checks (estimated ~800–1,000 additional calls). One model runs locally at zero marginal cost; the closed API model carries roughly 5,300–5,800 calls, comfortably inside a $20 budget cap at current mini/flash-tier pricing.

Timeline (12 Sep – 10 Nov):

- **Sep 12–19:** freeze literature review, hypotheses and analysis plan; finalize instruction paraphrases and message variants; set up local model runtime + API keys.
- **Sep 20–26:** build the 110-cell scenario bank; approx. 10-peer stimulus-quality check; pilot run on a small cell subset for ceiling/floor checks.
- **Sep 27–Oct 3:** finalize scenario bank and logging/checkpointing pipeline — Phase 1 (literature review) complete, mid-sem checkpoint.
- **Oct 4–17:** full data collection, both models (approx. 11,400 calls incl. checks), with buffer for refusal/parse-failure reruns.
- **Oct 18–24:** data cleaning; manipulation-check confusion matrices; belief-check covariate prep.
- **Oct 25–31:** mixed-effects models (S1–S3), Fehr–Schmidt fit, qualitative coding.
- **Nov 1–7:** write-up, figures, 0.7-temperature robustness subset if time allows, rehearse presentation.
- **Nov 10:** in-class presentation.

If data collection runs behind schedule, sub-experiments are completed in priority order S1, then S2, then S3, since S1 alone (with belief checks and the Fehr–Schmidt fit) is a complete test of the core hypotheses H1 and H3.

Limitations: hypothetical stakes; sensitivity to prompt wording; single-shot rather than repeated play; possible strategic information contained within emotional messages (partly addressed by the belief check); non-independence of repeated LLM generations; 2-model coverage; no access to any internal affective state. Null, ceiling/floor, or failed-manipulation results are reported as such, not interpreted as evidence for the hypotheses.

References

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